Microcornea–Myopic Chorioretinal Atrophy with Telecanthus (MMCAT): A Comprehensive Disease Report

Disease: Microcornea–Myopic Chorioretinal Atrophy (MMCAT syndrome) **Causal gene: ADAMTS18** (A Disintegrin And Metalloproteinase with Thrombospondin type 1 motifs, member 18) **OMIM:** 615458 · **MONDO:** 0014195 · **Category:** Mendelian, autosomal recessive

Summary

Microcornea–Myopic Chorioretinal Atrophy with Telecanthus (**MMCAT syndrome**; OMIM 615458; MONDO:0014195) is a rare, autosomal-recessive developmental eye disorder caused by **biallelic loss-of-function mutations in ADAMTS18**, a secreted zinc metalloproteinase located on chromosome 16q23.1 (HGNC:17110). Despite the disease name describing only clinical features, the defining molecular lesion is in ADAMTS18 — not COL18A1 or any myopia-susceptibility locus — established by autozygosity mapping followed by exome sequencing in consanguineous Saudi Arabian families (PMID: 23818446). ADAMTS18 is strongly expressed in the developing lens and retina, where it remodels the extracellular matrix (ECM) by cleaving fibronectin and regulating basement-membrane maturation. Its loss disrupts ocular ECM integrity during a critical window of early eye development.

The core clinical phenotype comprises **microcornea** (horizontal corneal diameter ≤10.5 mm), **high axial myopia** (spherical equivalent −6.00 to −13.75 D; axial length 25.6–26.8 mm) with **progressive myopic chorioretinal atrophy**, **telecanthus**, and **posteriorly-rotated ears**. The allelic spectrum of ADAMTS18 disease is broad and overlaps with **Knobloch syndrome**, encompassing childhood cataract, ectopia lentis, rhegmatogenous retinal detachment, cone–rod dystrophy, developmental glaucoma, and anterior-segment dysgenesis. The mechanistic basis is strongly supported by an *Adamts18*-knockout mouse that shows a **100%-penetrant eye defect caused by leakage of lens material through the lens capsule at embryonic day 13.5**, directly modeling the human lens/anterior-segment developmental lesion (PMID: 27638769).

MMCAT is ultra-rare, reported mainly in small consanguineous pedigrees with complete ocular penetrance but variable expressivity. There is **no curative or ADAMTS18-specific therapy**; management is supportive ophthalmic care (refractive correction, monitoring for retinal detachment, cataract and glaucoma treatment) plus genetic counseling. Life expectancy is normal; morbidity is primarily vision-threatening. This report synthesizes seven confirmed findings from 21 reviewed papers across all 15 requested disease-characteristic sections.

Key Findings

Finding 1 — ADAMTS18 biallelic mutations cause MMCAT syndrome

MMCAT was genetically defined by Aldahmesh and colleagues, who mapped the disorder to a single autozygous locus on **chromosome 16q23.1**. Exome sequencing in the index consanguineous family revealed a **homozygous missense change in ADAMTS18**, and targeted direct sequencing of four additional probands with the same phenotype revealed three further homozygous changes, **including two nonsense mutations**. The autozygomes of all probands overlapped at 16q23.1, cementing ADAMTS18 as causal.

> *"we show that MMCAT maps to a single autozygous locus on chr16q23.1 in which exome sequencing revealed a homozygous missense change in ADAMTS18"* (PMID: 23818446)
>
> *"Our data suggest that ADAMTS18 plays an essential role in early eye development and that mutations therein cause a distinct eye phenotype that is mainly characterized by microcornea and myopia."* (PMID: 23818446)

The presence of nonsense alleles among the causal mutations indicates a **loss-of-function mechanism**. Inheritance is autosomal recessive (OMIM 615458; MONDO:0014195).

Finding 2 — ADAMTS18 disease forms a phenotypic spectrum overlapping Knobloch syndrome

ADAMTS18 was independently identified as a gene mutated in **Knobloch syndrome** (classically a COL18A1 disorder) through combined autozygosity and exome analysis (PMID: 21862674). This established allelic and phenotypic overlap between MMCAT and Knobloch syndrome.

> *"the study identified ADAMTS18 as the only gene carrying a homozygous protein altering mutation"* (PMID: 21862674)

The ocular phenotype was subsequently expanded across three families carrying novel homozygous mutations — **c.1067T>A (p.L356*), c.2159G>C (p.C720S), and c.1952G>A (p.R651Q)** — adding bilateral microcornea, ectopic pupils, childhood cataract, ectopia lentis, rhegmatogenous retinal detachment, and cone–rod dystrophy to the spectrum (PMID: 24874986).

> *"In a second family, 2 brothers showed bilateral microcornea, childhood cataract, ectopia lentis, rhegmatogenous retinal detachment, and cone-rod dystrophy."* (PMID: 24874986)

Further widening the spectrum: a compound-heterozygous 37-year-old male showed **anterior-segment dysgenesis, retinal dystrophy, nasolacrimal abnormalities, altered lens dimensions/shape, and preserved visual acuity at older age** (PMID: 39902400); and novel compound-heterozygous mutations **c.3436C>T (p.R1146W) and c.1454T>G (p.F485C)** produced **bilateral developmental glaucoma with microcornea** in twins (PMID: 38199772).

Finding 3 — ADAMTS18 is a secreted zinc metalloproteinase that remodels ECM; disease arises from ECM dysregulation during eye development

ADAMTS18 is a member of the ADAMTS superfamily of 19 secreted zinc metalloproteinases that assemble and degrade ECM during development, morphogenesis, and tissue repair. Mechanistically, ADAMTS18 **proteolytically cleaves fibronectin (FN)** at two defined sites and **negatively regulates FN fibrillogenesis**; its loss causes FN accumulation, altered epithelial branching, and reduced endothelial sprouting (PMID: 41135673).

> *"we identified two sites in FN that are proteolytically cleaved by ADAMTS18"* (PMID: 41135673)

ADAMTS18 is strongly expressed in the developing eye:

> *"It was also shown that Adamts18 is expressed in the lens and retina in the developing murine eye."* (PMID: 21862674)

Parallel roles reinforce the ECM mechanism: ADAMTS18 deficiency alters mammary ECM components (laminin-α5, fibronectin, collagen I; PMID: 38287441) and disrupts glomerular basement-membrane isoform maturation and glomerular filtration-barrier integrity (PMID: 39879696). A thrombin-cleaved C-terminal fragment has a distinct activity inducing oxidative platelet fragmentation (PMID: 19218546).

Finding 4 — Adamts18-knockout mouse recapitulates the lens-capsule lesion with 100% penetrance at E13.5

Ataca et al. generated *Adamts18*-null mice (C57Bl6/Ola background) that are born at normal Mendelian ratios and viable, with transient growth delay. Histology revealed the defining ocular lesion:

> *"a 100% penetrant eye defect resulting from leakage of lens material through the lens capsule occurring at embryonic day (E)13.5, when the lens grows rapidly"* (PMID: 27638769)

Additional knockout defects include altered bronchiolar branching in lung and reproductive-tract anomalies (≈50% of females infertile from vaginal septum/imperforate vagina; increased ovarian rete). A review confirms broad developmental roles:

> *"ADAMTS18 is essential for the morphology and organogenesis of several epithelial organs (e.g., lung, kidney, breast, salivary glands, and lacrimal glands), vascular and neuronal systems, adipose tissue, and reproductive tracts"* (PMID: 36632911)

The lens-capsule leakage during the rapid-growth phase is the mechanistic linchpin connecting ADAMTS18 ECM dysfunction to microcornea, cataract, ectopia lentis, and secondary high myopia.

Finding 5 — ADAMTS18 is pleiotropic beyond the eye (bone, platelet/thrombosis, tumor suppression)

A genome-wide association study (n = 5,925–9,828 with replication) identified ADAMTS18 as a bone-mineral-density candidate gene (meta-analysis p = 2.56×10^{-5} to 2.13×10^{-8}) with hip-fracture association in a Chinese cohort (PMID: 19249006).

> *"The GWAS detected two BMD candidate genes, ADAMTS18 (ADAM metallopeptidase with thrombospondin type 1 motif, 18) and TGFBR3"* (PMID: 19249006)
>
> *"Mutations of ADAMTS-18 have been linked to abnormal early eye development and reduced bone mineral density."* (PMID: 24896365)

Additional pleiotropy: ADAMTS18 acts as a functional **tumor suppressor** (its deficiency increases HER2+ mammary tumorigenesis and metastasis; PMID: 38287441), and a thrombin-cleaved C-terminal fragment regulates platelet thrombus clearance (PMID: 19218546). These roles are relevant to differential diagnosis and potential modifier considerations but are not part of the core MMCAT ocular syndrome.

Finding 6 — Quantitative MMCAT clinical phenotype

The original clinical delineation (Khan 2012) described **5 boys (4–10 years) from 4 consanguineous/endogamous Saudi families**. Microcornea was defined as horizontal corneal diameter ≤10.5 mm (range 9.8–10.5 mm); anterior chambers were **not** shallow (2.66–3.01 mm), distinguishing MMCAT from nanophthalmos/microphthalmos with crowded anterior segment.

> *"myopic chorioretinal atrophy was severe and associated with a cycloplegic spherical equivalent at least -6.00 diopters (-6.00 to -13.75) and an axial length of at least 25.63 mm (25.63-26.84)"* (PMID: 22686506)
>
> *"All five affected children had telecanthus and posteriorly-rotated ears, which were not present in unaffected relatives."* (PMID: 22686506)

Expressivity was variable: in 2 affected siblings, findings were milder (SE −0.50 to −2.50 D; axial length 22.42–23.69 mm). Best-corrected visual acuity was **20/60 or better in children older than 4 years**, indicating preserved central vision in childhood. One father was also affected (pseudodominant transmission). Congenital nasolacrimal duct obstruction and dry eye have also been reported (PMID: 39902400).

Finding 7 — Ultra-rare AR disorder with complete ocular penetrance, variable expressivity, strong consanguinity, normal life expectancy

Inheritance is autosomal recessive with possible pseudodominant transmission in consanguineous/endogamous pedigrees (PMID: 23818446; PMID: 22686506).

> *"Five boys (4-10 years old) from four consanguineous and/or endogamous Saudi Arabian families were identified."* (PMID: 22686506)
>
> *"Consistent with the presumed pseudodominant inheritance in one of the original families"* (PMID: 23818446)

Ocular penetrance appears complete (100% in the knockout mouse); expressivity is variable. No published prevalence or incidence figures exist; the condition is ultra-rare, reported in small consanguineous Saudi series and isolated cases worldwide (UK, China, USA). The disease is **not life-limiting**; morbidity is primarily visual. Management is supportive, with no curative or ADAMTS18-specific therapy and no registered clinical trials identified.

Report by Section

1. Disease Information

MMCAT is a rare autosomal-recessive developmental eye syndrome. **Key identifiers:** OMIM 615458; MONDO:0014195; causal gene ADAMTS18 (HGNC:17110; chromosome 16q23.1). Orphanet lists it under ADAMTS18-related disease; discrete ICD-10/ICD-11 and MeSH codes specific to MMCAT are not established (it falls under broad categories for microcornea, high myopia, and chorioretinal atrophy). **Synonyms:** MMCAT syndrome; microcornea–myopic chorioretinal atrophy–telecanthus syndrome. Information is derived from **aggregated disease-level resources and small clinical case series**, not large EHR cohorts.

2. Etiology

Primary cause: monogenic — biallelic (homozygous or compound-heterozygous) loss-of-function mutations in ADAMTS18 (PMID: 23818446). **Genetic risk factors:** consanguinity dramatically increases risk by producing homozygosity for rare recessive alleles; all original families were consanguineous/endogamous. No environmental, infectious, toxic, or lifestyle causes are implicated. No protective alleles or gene–environment interactions have been reported. The axial elongation (high myopia) is a developmental consequence of ECM failure rather than the classic environmental/behavioral myopia of the general population.

3. Phenotypes

Phenotype	Type	HPO term (suggested)	Onset	Frequency / Notes
Microcornea (≤10.5 mm)	Physical/ clinical sign	HP:0000482	Congenital	Core feature, ~all patients
High myopia (−6 to −13.75 D)	Clinical sign	HP:0011003	Childhood, progressive	Core; variable severity
Myopic chorioretinal atrophy	Clinical sign	HP:0040049 / chorioretinal atrophy	Childhood, progressive	Core, progressive
Increased axial length (25.6–26.8 mm)	Biometric/ lab	HP:0007843	Childhood	Core
Telecanthus	Physical sign	HP:0000506	Congenital	5/5 original patients
Posteriorly-rotated ears	Physical sign	HP:0000358	Congenital	5/5 original patients
Cataract (childhood)	Clinical sign	HP:0000518	Childhood	Spectrum/ Knobloch overlap
Ectopia lentis	Clinical sign	HP:0001083	Childhood	Spectrum
Rhegmatogenous retinal detachment	Clinical sign	HP:0012230	Childhood/ adult	Spectrum; vision-threatening
Cone–rod dystrophy	Clinical sign	HP:0000548	Childhood	Spectrum
Developmental glaucoma	Clinical sign	HP:0001087 / HP:0000501	Childhood	Reported variant family
Nasolacrimal duct obstruction	Clinical sign	HP:0000579	Congenital	Reported

Quality-of-life impact: Central vision is often preserved in childhood (BCVA ≥20/60), but progressive chorioretinal atrophy, high myopia, and risk of retinal detachment, cataract, and glaucoma threaten long-term vision and affect education, mobility, and independence.

4. Genetic / Molecular Information

Causal gene: ADAMTS18 (HGNC:17110; OMIM gene 607512; NCBI Gene 170692; UniProt Q8TE60; chromosome 16q23.1). **Reported pathogenic variants** include a homozygous missense (index MMCAT family) plus nonsense alleles (PMID: 23818446); c.1067T>A (p.L356*, nonsense), c.2159G>C (p.C720S, missense), c.1952G>A (p.R651Q, missense) (PMID: 24874986); and compound-heterozygous c.3436C>T (p.R1146W) + c.1454T>G (p.F485C) (PMID: 38199772). **Variant classes:** missense and nonsense predominate (loss-of-function). **Classification:** pathogenic/likely pathogenic per ACMG. **Allele frequency:** all are rare in gnomAD (consistent with ultra-rare recessive disease). **Origin:** germline. **Functional consequence:** loss of function (reduced/abolished metalloproteinase activity → ECM dysregulation). No established modifier genes or disease-specific epigenetic or chromosomal abnormalities; however, ADAMTS18 methylation is a documented regulatory mechanism in fibrosis contexts (PMID: 37812341).

5. Environmental Information

No environmental, toxic, occupational, lifestyle, or infectious factors are implicated in MMCAT. It is a purely monogenic developmental disorder. **CHEBI relevance:** zinc (CHEBI:29105) as the catalytic cofactor of the metalloproteinase.

6. Mechanism / Pathophysiology

Ordered causal chain:

1. Biallelic loss-of-function mutation in **ADAMTS18 → loss of secreted zinc-metalloproteinase activity** (demonstrated: nonsense alleles, PMID: 23818446).
2. Loss of ADAMTS18 protease activity → **failure to cleave fibronectin and properly remodel/mature basement-membrane ECM** (demonstrated in vitro: FN cleavage at two sites, PMID: 41135673; GBM maturation, PMID: 39879696).
3. ECM/basement-membrane dysregulation in the developing eye (ADAMTS18 expressed in lens and retina, PMID: 21862674) → **structural weakness of the lens capsule** during rapid lens growth.
4. Lens-capsule weakness → **leakage of lens material through the capsule at E13.5** (demonstrated in knockout mouse, 100% penetrant, PMID: 27638769).

5. Disrupted anterior-segment/lens development → **microcornea, cataract, ectopia lentis** (inferred from mouse lesion + human phenotype).
6. Abnormal ocular ECM and growth regulation → **dysregulated axial elongation → high axial myopia** (inferred mechanistic link from biometry, PMID: 22686506).
7. High myopia + abnormal retinal/choroidal ECM → **progressive myopic chorioretinal atrophy**, with downstream risk of **rhegmatogenous retinal detachment, cone–rod dystrophy, and glaucoma** (clinical, PMID: 24874986; PMID: 38199772).
8. Branch: parallel ECM roles in periocular/craniofacial development → **telecanthus, posteriorly-rotated ears, nasolacrimal duct obstruction** (inferred).

```
ADAMTS18 biallelic LoF
        │ (loss of metalloprotease activity)
        ▼
Failure to cleave fibronectin / mature basement-membrane ECM
        │
        ▼
Lens-capsule ECM weakness (lens + retina expression)
        │
        ├──► Lens-capsule leakage at E13.5 ──► microcornea, cataract, ectopia lentis
        │
        ├──► dysregulated axial growth ──► high axial myopia ──► myopic chorioretinal atrophy
        │                                                          └──► retinal detachment, CRD,
        │
        └──► periocular/craniofacial ECM defect ──► telecanthus, posteriorly-rotated ears, NLDO
```

Upstream: ADAMTS18 loss and FN/ECM dysregulation. **Downstream:** lens, corneal, retinal, and craniofacial malformations. **GO terms:** extracellular matrix organization (GO:0030198), extracellular matrix disassembly (GO:0022617), metalloendopeptidase activity (GO:0004222), eye development (GO:0001654), camera-type eye development (GO:0043010), lens development in camera-type eye (GO:0002088). **Cell types (CL):** lens fiber cell (CL:0000362), corneal epithelial cell, retinal pigment epithelial cell (CL:0002586), photoreceptor cell (CL:0000210), vascular endothelial cell (CL:0000115). **Protein dysfunction:** loss of function via truncation/ missense impairing catalysis or secretion.

7. Anatomical Structures Affected

Organ level (primary): eye (UBERON:0000970) — cornea (UBERON:0000964), lens (UBERON:0000965), retina (UBERON:0000966), choroid (UBERON:0001776), anterior chamber, iris, ciliary body. **Secondary/periocular:** eyelid/canthus region (telecanthus), external ear (UBERON:0001690; posteriorly-rotated ears), nasolacrimal duct (UBERON:0002392). **Body**

systems: primarily the **visual/sensory system**, with craniofacial involvement. **Tissue level:** epithelial (lens, corneal), connective/ECM (lens capsule — a specialized basement membrane), neural retina. **Subcellular (GO CC):** extracellular region/matrix (GO:0005615, GO:0031012), basement membrane (GO:0005604). **Lateralization:** bilateral.

8. Temporal Development

Onset: congenital — microcornea, telecanthus, and ear anomalies are present at birth; the underlying lens-capsule lesion arises at E13.5 in the mouse model. **Progression:** high myopia and chorioretinal atrophy are **progressive** through childhood; central vision is relatively preserved early (BCVA ≥20/60 after age 4). **Disease course:** chronic, lifelong, slowly progressive. **Critical period:** early embryonic eye development (lens rapid-growth phase) is the window of vulnerability — no intervention window currently exists, but it defines when ECM integrity matters most.

9. Inheritance and Population

Inheritance: autosomal recessive; pseudodominant transmission possible in consanguineous/endogamous pedigrees (PMID: 23818446). **Penetrance:** ocular penetrance appears complete (100% in knockout mouse). **Expressivity:** variable (severe vs mild myopia within/between families). **Epidemiology:** ultra-rare; no published prevalence/incidence. **Consanguinity:** strong association; reported primarily in Saudi Arabian consanguineous families, with isolated cases in the UK, China, and USA. **Founder effect:** plausible in the endogamous Saudi population but not formally quantified. **Sex ratio:** original series were all male, but this reflects small-family ascertainment, not sex linkage (ADAMTS18 is autosomal). No genetic anticipation (not a repeat-expansion disorder).

10. Diagnostics

Clinical/ophthalmic: slit-lamp examination (microcornea, cataract, ectopia lentis), **corneal diameter measurement** (≤10.5 mm diagnostic), **cycloplegic refraction** (high myopia), **A-scan/optical biometry** (axial length ≥25.6 mm), **dilated fundoscopy and OCT** (chorioretinal atrophy, retinal nerve-fiber-layer/ganglion-cell thinning in glaucoma cases; PMID: 38199772), **tonometry/gonioscopy** (glaucoma screening), and craniofacial assessment (telecanthus, ear position). **Genetic testing is definitive:** single-gene ADAMTS18 sequencing, targeted retinal-dystrophy/anterior-segment gene panels, or whole-exome sequencing; **autozygosity mapping** is powerful in consanguineous families (PMID: 23818446; PMID: 21862674). **Differential diagnosis:** Knobloch syndrome (COL18A1; look for occipital encephalocele), nanophthalmos/

posterior microphthalmos (shallow anterior chamber — notably NOT present in MMCAT), Marfan/other ectopia-lentis syndromes, isolated high myopia, and anterior-segment dysgenesis of other genetic causes. No biochemical biomarker or newborn-screening assay exists.

11. Outcome / Prognosis

Survival/mortality: normal life expectancy; not life-limiting. **Morbidity:** primarily ophthalmic — progressive vision loss from chorioretinal atrophy, with risk of retinal detachment, cataract, and glaucoma. **Recovery:** developmental malformations (microcornea, telecanthus) are non-reversible; refractive error is correctable; some complications (cataract, glaucoma, detachment) are treatable surgically/medically. **Prognostic factors:** severity of myopia/axial length and onset of retinal complications predict visual outcome. Preserved central acuity in childhood is a favorable early sign, but long-term chorioretinal atrophy and complications determine ultimate prognosis.

12. Treatment

There is **no curative or ADAMTS18-specific therapy and no registered clinical trial**. Management is **supportive ophthalmic care**: - **Refractive correction** of high myopia (spectacles/contact lenses) — NCIT: corrective lens therapy. - **Cataract surgery** when visually significant — NCIT: cataract extraction/lens surgery. - **Glaucoma management** (IOP-lowering medications, surgery) in affected individuals (PMID: 38199772). - **Retinal-detachment surveillance and surgical repair** (vitrectomy/scleral buckle) given rhegmatogenous risk. - **Management of dry eye and nasolacrimal duct obstruction** (lubrication, probing/surgery) (PMID: 39902400). - **Low-vision rehabilitation** and **genetic counseling**.

No pharmacogenomic, gene-therapy, cell-therapy, or RNA-based therapy exists for this gene/disease. The pleiotropic tumor-suppressor and bone roles of ADAMTS18 are not clinically actionable in MMCAT.

13. Prevention

Primary prevention: none for disease occurrence (monogenic/developmental); **genetic counseling** for consanguineous families and **carrier screening/cascade testing** of at-risk relatives are the main preventive tools. **Reproductive options:** preimplantation genetic testing (PGT) and prenatal diagnosis are feasible once the familial ADAMTS18 variants are known. **Secondary/tertiary prevention:** early ophthalmic surveillance to detect and treat cataract, glaucoma, and retinal detachment, preventing avoidable vision loss. No vaccination, behavioral, or public-health intervention applies.

14. Other Species / Natural Disease

Model-validated ortholog: mouse *Adamts18* (NCBI Gene 71957). The gene is evolutionarily conserved across vertebrates. **Natural disease in other species:** no well-documented naturally occurring ADAMTS18 ocular disease in companion animals or wildlife is recorded in OMIA at the level reported for humans; the disease model is engineered (knockout). No zoonotic potential (non-infectious genetic disorder). **NCBI Taxon:** human (9606), mouse (10090).

15. Model Organisms

The principal model is the **Adamts18-knockout mouse** (C57Bl6/Ola), a constitutive knockout (PMID: 27638769). **Phenotype recapitulation:** excellent for the ocular developmental lesion — 100%-penetrant lens-capsule leakage at E13.5 models the human microcornea/lens/anterior-segment phenotype. It additionally reproduces pleiotropic organ defects (lung branching, reproductive tract, salivary/lacrimal glands, kidney GBM, preputial gland, vasculature) as cataloged across multiple studies (PMID: 36632911; PMID: 34388417; PMID: 34323105; PMID: 34271244; PMID: 34189436; PMID: 39879696). **Limitations:** the mouse does not fully capture adult human chorioretinal atrophy progression or the telecanthus/ear craniofacial features, and myopia is difficult to phenocopy. **Applications:** studying ECM/fibronectin regulation, lens-capsule biology, and organogenesis. **Resources:** MGI for mouse allele/strain data.

Mechanistic Model / Interpretation

MMCAT is fundamentally a **developmental ECM disorder of the eye**. The unifying model is that ADAMTS18, a secreted zinc metalloproteinase strongly expressed in the embryonic lens and retina, maintains the structural integrity of ocular ECM — most critically the **lens capsule**, a specialized basement membrane — during the period of rapid lens growth. When both ADAMTS18 alleles are lost, fibronectin is not properly cleaved and basement-membrane macromolecules fail to mature, weakening the lens capsule. In the mouse this produces frank leakage of lens material at E13.5; in humans the homologous failure manifests as microcornea, cataract/ectopia lentis, abnormal axial elongation (high myopia), and progressive myopic chorioretinal atrophy, with periocular/craniofacial ECM involvement producing telecanthus, posteriorly-rotated ears, and nasolacrimal duct anomalies. The disease sits on an **allelic/phenotypic continuum with Knobloch syndrome**, explaining why cone–rod dystrophy, retinal detachment, and glaucoma appear across different ADAMTS18 families. The

convergence of human genetics (nonsense alleles), biochemistry (fibronectin cleavage), expression data (lens/retina), and a 100%-penetrant knockout lens-capsule phenotype makes this one of the better-supported ECM–eye-development mechanisms among rare ophthalmic disorders.

Evidence Base

PMID	Title (abbrev.)	Role in this report
23818446	*MMCAT caused by mutations in ADAMTS18*	Establishes causal gene; locus 16q23.1; nonsense alleles → LoF
22686506	*Microcornea with myopic chorioretinal atrophy, telecanthus…*	Original quantitative clinical delineation
21862674	*ADAMTS18 in Knobloch syndrome*	Allelic overlap; lens/retina expression
24874986	*Expansion of ocular phenotypes in ADAMTS18*	Phenotypic spectrum; named variants
38199772	*Developmental glaucoma with microcornea, novel ADAMTS18*	Glaucoma phenotype; compound-het variants
39902400	*Expanded genotype/phenotype in ADAMTS18*	Adult compound-het; anterior-segment dysgenesis, NLDO
41135673	*ADAMTS18 cleaves fibronectin*	Biochemical mechanism
27638769	*Adamts18 deletion developmental defects*	Knockout mouse; 100% penetrant lens-capsule leakage
36632911	*Secreted protease ADAMTS18 in development and disease*	Review of pleiotropic organogenesis
39879696	*ADAMTS18 and glomerular basement membrane*	Basement-membrane maturation role
38287441	*ADAMTS18 deficiency and HER2+ tumorigenesis*	ECM components; tumor-suppressor pleiotropy
19249006	*GWAS ADAMTS18 and bone mass*	Bone-density pleiotropy
19218546	*C-terminal ADAMTS-18 fragment and platelets*	Hematologic pleiotropy
24896365	*ADAMTS-18: multiple functions (review)*	Links eye + bone phenotypes
37812341	*Curcumin reverses ADAMTS18 methylation*	Epigenetic regulation context

Evidence types span **human clinical** (case series, genetics), **model organism** (knockout mice), **in vitro** (fibronectin cleavage, cell models), and **computational/GWAS** (population genetics). All core claims are supported by verified abstract quotes.

Limitations and Knowledge Gaps

- **No epidemiological data:** prevalence, incidence, and carrier frequency are unknown; the disease is defined by small consanguineous series, creating ascertainment bias (e.g., apparent male predominance despite autosomal inheritance).
- **Genotype–phenotype correlation is immature:** it is unclear why some alleles yield mild MMCAT while others produce Knobloch-like disease, glaucoma, or cone–rod dystrophy. Missense vs truncating effects on residual protease activity are not systematically characterized.
- **The axial myopia mechanism is inferred, not demonstrated** — the causal link from ADAMTS18 ECM dysfunction to abnormal axial elongation has not been directly shown mechanistically.
- **Mouse model incompletely phenocopies humans:** telecanthus, ear position, and progressive chorioretinal atrophy are not clearly modeled.
- **No natural animal disease** or dedicated therapeutic development; no biomarkers for progression.
- **Modifier genes and epigenetic contributions** to variable expressivity are unexplored.

Proposed Follow-up Experiments / Actions

1. **Variant–function assays:** express each reported ADAMTS18 missense/nonsense allele and quantify secretion and fibronectin-cleavage activity to build a genotype–activity–phenotype map.
2. **Conditional/lens-specific knockouts and knock-in of human alleles** in mice to directly test the axial-elongation and chorioretinal-atrophy mechanisms, with longitudinal ocular biometry.
3. **International patient registry** and reverse-phenotyping across ophthalmic genetics cohorts (e.g., retinal-dystrophy panels) to estimate prevalence and refine the phenotypic spectrum.

4. **Deep ocular ECM phenotyping** (lens capsule, Bruch's membrane, choroid) by immunohistochemistry/proteomics in model tissue to identify downstream ECM substrates beyond fibronectin.
5. **Natural-history study with OCT/biometry** to define progression rates of chorioretinal atrophy and complication risk, informing surveillance guidelines.
6. **Preclinical proof-of-concept** for ECM-stabilizing or protease-replacement/gene-replacement strategies, given that loss-of-function is the mechanism and ADAMTS18 is a secreted protein (theoretically amenable to supplementation).
